INCH-POUND

MIL-C-85049/8A
30 November 1990
SUPERSEDING
MIL-C-85049/8
16 January 1980
MS3188B
10 May 1982

MILITARY SPECIFICATION SHEET

CONNECTOR ACCESSORIES, ELECTRICAL, BACKSHELL, ENVIRONMENTAL, CABLE SEALING, 90°, SHIELD TERMINATION, CATEGORY 1A (FOR MIL-C-5015 CRIMP, MIL-C-26482 SERIES 2, MIL-C-81703 SERIES 3, AND MIL-C-83723 SERIES III CONNECTORS)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-C-85049.

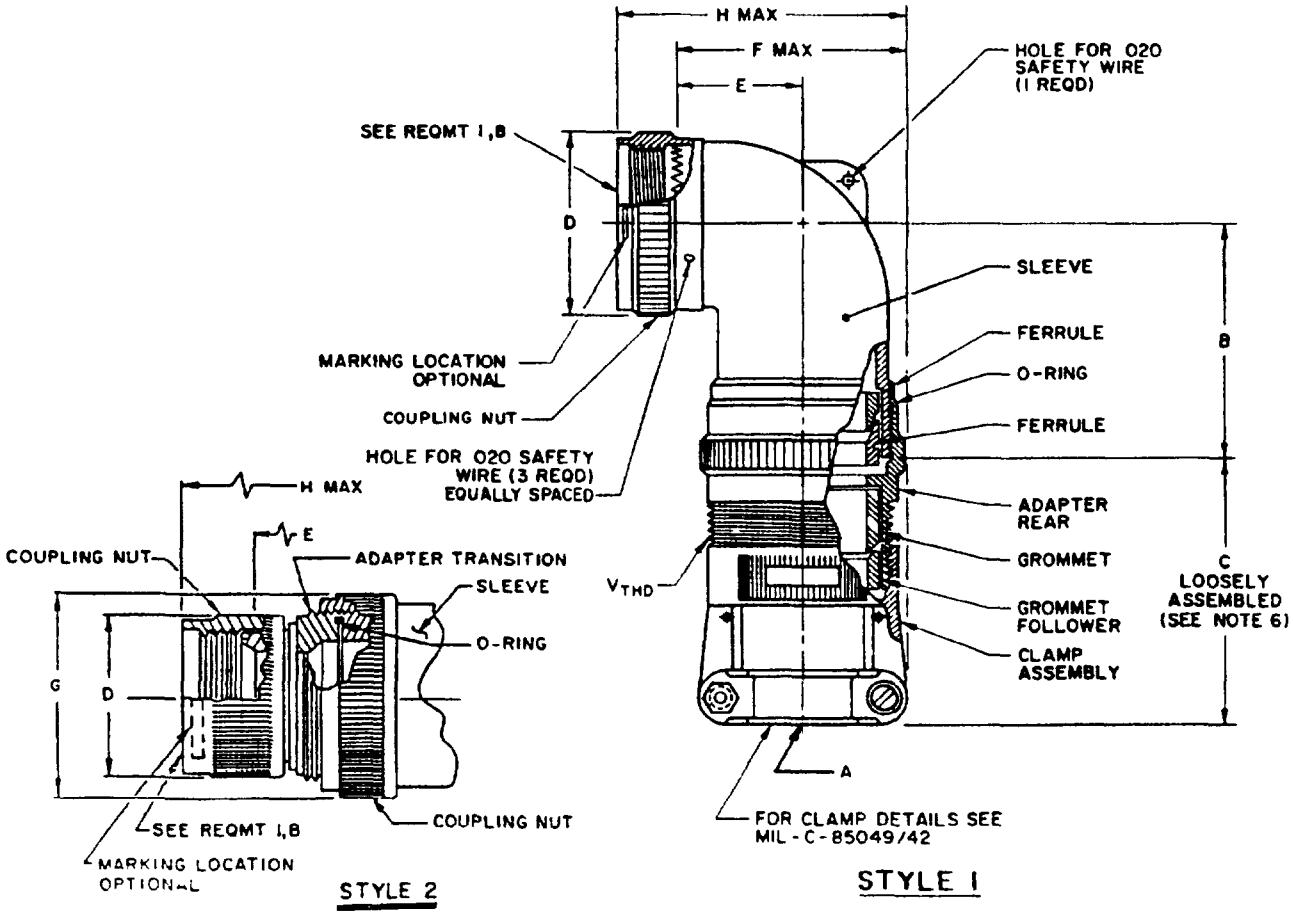


FIGURE 1. Configuration and dimensions.

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TABLE I. Dimensions.

Dash No.	Fits Shell of Connector				Clamp Size	A (see note 6) Wire Bundle Accommodation Range		Style	B ±.062	C Ref	D Dia Max
	MIL-C-81703 Ser 3	MIL-C-26482 Ser 2	MIL-C-5015 Crimp	MIL-C-83723 Ser III		Min	Max				
1	3				4	.125	.250	1	1.800	1.544	.669
2					6	.250	.437	2	1.320	1.544	.669
3		8	8S	8	4	.125	.250	1	1.200	1.544	.617
4		10	10S, 10SL	10	4	.125	.312	1	1.920	1.544	.734
5					6	.250	.375	1	1.320	1.544	.734
6	7	12	12,12S	12	4	.125	.312	1	1.940	1.544	.858
7					6	.250	.437	1	1.940	1.544	.858
8					10	.350	.500	1	1.335	1.844	.858
9	12	14	14,14S	14	6	.250	.437	1	2.010	1.544	.984
10					10	.350	.575	1	1.655	1.844	.984

Dash No.	E ±.125	V Thread Class 2A	F Max	G Dia Max	H Max
1	.636	.625-24 UNEF	1.18	-	1.57
2	1.386	.750-20 UNEF	2.02	.858	2.41
3	.541	.625-24 UNEF	1.09	-	1.48
4	.636	.625-24 UNEF	1.18	-	1.57
5	.636	.750-20 UNEF	1.27	-	1.66
6	.641	.625-24 UNEF	1.18	-	1.57
7	.641	.750-20 UNEF	1.28	-	1.67
8	.641	1.000-20 UNEF	1.37	-	1.76
9	.741	.750-20 UNEF	1.38	-	1.77
10	.741	1.000-20 UNEF	1.47	-	1.86

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TABLE I. Dimensions (continued).

Dash No.	Fits Shell of Connector				Clamp Size	A (see note 6) Wire Bundle Accommodation Range		Style	B ±.062	C Ref	D Dia Max
	MIL-C-81703 Ser 3	MIL-C-26482 Ser 2	MIL-C-5015 Crimp	MIL-C-83723 Ser III		Min	Max				
11	19	16	16,16S	16	6	.250	.437	1	2.100	1.544	1.112
12					12	.500	.700	1	1.745	1.916	1.112
13	27	18	18	18	10	.350	.625	1	2.270	1.844	1.218
14					16	.625	.779	1	1.920	2.000	1.218
15	37	20	20	20	10	.350	.625	1	2.270	1.844	1.345
16					16	.625	.904	1	1.920	2.000	1.345
17		22	22	22	12	.500	.750	1	2.380	1.918	1.468
18					20	.875	1.029	1	2.025	2.230	1.468
19		24	24	24	12	.500	.750	1	2.380	1.916	1.593
20					20	.875	1.144	1	2.025	2.230	1.593

Dash No.	E ±.125	V Thread Class 2A	F Max	G Dia Max	H Max
11	.926	.750-20 UNEF	1.59	-	1.98
12	.926	1.1875-18 UNEF	1.77	-	2.16
13	1.016	1.000-20 UNEF	1.80	-	2.19
14	1.016	1.4375-18 UNEF	1.97	-	2.36
15	1.016	1.000-20 UNEF	1.80	-	2.19
16	1.016	1.4375-18 UNEF	1.97	-	2.36
17	1.166	1.1875-18 UNEF	2.11	-	2.50
18	1.166	1.750-18 UNS	2.29	-	2.68
19	1.166	1.1875-18 UNEF	2.11	-	2.50
20	1.166	1.750-18 UNS	2.29	-	2.68

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TABLE I. Dimensions (continued).

Dash No.	Fits Shell of Connector				Clamp Size	A (see note 6) Wire Bundle Accommodation Range		Style	B ±.062	C Ref	D Dia Max
	MIL-C-81703 Ser 3	MIL-C-26482 Ser 2	MIL-C-5015 Crimp	MIL-C-83723 Ser III		Min	Max				
21			28		16	.625	.937	1	2.550	2.000	1.969
22					24	1.000	1.375	1	2.195	2.024	1.969
23			32		16	.625	.937	1	2.800	2.000	2.219
24					20	.875	1.250	1	2.800	2.230	2.219
25					28	1.250	1.625	1	2.445	2.550	2.219
26			36		16	.625	.937	1	2.770	2.000	2.469
27					24	1.000	1.375	1	2.770	2.024	2.469
28					32	1.437	1.840	1	2.415	2.600	2.469
29			40		16	.625	.937	1	2.770	2.000	2.719
30					24	1.000	1.375	1	2.770	2.024	2.719

Dash No.	E ±.125	V Thread Class 2A	F Max	G Dia Max	H Max
21	1.266	1.4375-18 UNEF	2.33	-	2.88
22	1.266	2.000-18 UNS	2.51	-	3.06
23	1.616	1.4375-18 UNEF	2.81	-	3.36
24	1.616	1.750-18 UNS	2.81	-	3.36
25	1.616	2.250-16 UN	3.07	-	3.62
26	1.816	1.4375-18 UNEF	3.13	-	3.68
27	1.816	2.000-18 UNS	3.13	-	3.68
28	1.816	2.500-16 UN	3.39	-	3.94
29	2.566	1.4375-18 UNEF	3.88	-	4.43
30	2.566	2.000-18 UNS	3.88	-	4.43

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TABLE I. Dimensions (continued).

Dash No.	Fits Shell of Connector				Clamp Size	A (see note 6) Wire Bundle Accommodation Range		Style	B ±.062	C Ref	D Dia Max
	MIL-C-81703 Ser 3	MIL-C-26482 Ser 2	MIL-C-5015 Crimp	MIL-C-83723 Ser III		Min	Max				
31			40		32	1.437	1.875	1	2.415	2.600	2.719
32			44		16	.625	.937	1	2.770	2.000	2.969
33					24	1.000	1.375	1	2.770	2.024	2.969
34					32	1.437	1.875	1	2.415	2.600	2.969
35			48		16	.625	.937	1	2.770	2.000	3.219
36					24	1.000	1.375	1	2.770	2.024	3.219
37					32	1.437	1.875	1	2.415	2.600	3.219
38	61				12	.500	.750	1	2.380	1.916	1.653
39					20	.875	1.184	1	2.025	2.230	1.653
40	19	16	16,16S	16	10	.350	.625	1	2.100	1.844	1.112

Dash No.	E ±.125	V Thread Class 2A	F Max	G Dia Max	H Max
31	2.566	2.500-16 UN	4.14	-	4.69
32	2.566	1.4375-18 UNEF	3.88	-	4.43
33	2.566	2.000-18 UNS	3.88	-	4.43
34	2.566	2.500-16 UN	4.14	-	4.69
35	2.566	1.4375-18 UNEF	3.88	-	4.43
36	2.566	2.000-18 UNS	3.88	-	4.43
37	2.566	2.500-16 UN	4.14	-	4.69
38	1.166	1.1875-18 UNEF	2.11	-	2.50
39	1.166	1.750-18 UNS	2.29	-	2.68
40	.926	1.000-20 UNEF	1.66	-	2.05

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TABLE I. Dimensions (continued).

Dash No.	Fits Shell of Connector				Clamp Size	A (see note 6) Wire Bundle Accommodation Range		Style	B ±.062	C Ref	D Dia Max
	MIL-C-81703 Ser 3	MIL-C-26482 Ser 2	MIL-C-5015 Crimp	MIL-C-83723 Ser III		Min	Max				
41	27	18	18	18	4	.125	.312	1	2.270	1.544	1.218
42					6	.250	.437	1	2.270	1.544	1.218
43	37	20	20	20	6	.250	.437	1	2.270	1.544	1.345
44		22	22	22	4	.125	.312	1	2.380	1.544	1.468
45					6	.250	.437	1	2.380	1.544	1.468
46		24	24	24	10	.350	.625	1	2.380	1.844	1.593
47			36		12	.500	.750	1	2.770	1.916	2.469
48			40		12	.500	.750	1	2.770	1.916	2.719
49		10	10S, 10SL	10	6	.250	.437	2	1.940	1.544	.734
50	12	14	14,14S	14	12	.500	.750	2	2.270	1.916	.984

Dash No.	E ±.125	V Thread Class 2A	F Max	G Dia Max	H Max
41	1.016	.625-24 UNEF	1.80	-	2.19
42	1.016	.750-20 UNEF	1.80	-	2.19
43	1.016	.750-20 UNEF	1.80	-	2.19
44	1.166	.625-24 UNEF	2.11	-	2.50
45	1.166	.750-20 UNEF	2.11	-	2.50
46	1.166	1.000-20 UNEF	2.11	-	2.50
47	1.816	1.1875-18 UNEF	3.13	-	3.68
48	1.816	1.1875-18 UNEF	3.88	-	4.43
49	1.391	.750-20 UNEF	2.03	.858	2.42
50	1.766	1.1875-18 UNEF	2.61	1.218	3.00

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TABLE I. Dimensions (continued).

Dash No.	Fits Shell of Connector				Clamp Size	A (see note 6) Wire Bundle Accommodation Range		Style	B ±.062	C Ref	D Dia Max
	MIL-C-81703 Ser 3	MIL-C-26482 Ser 2	MIL-C-5015 Crimp	MIL-C-83723 Ser III		Min	Max				
51	19	16	16,16S	16	16	.625	.937	2	2.380	2.000	1.112
52	27	18	18	18	12	.500	.750	1	2.270	1.916	1.218
53	61				16	.625	.937	1	2.025	2.000	1.653
54	37	20	20	20	12	.500	.750	1	2.270	1.916	1.345
55					20	.875	1.250	2	2.550	2.230	1.345
56		22	22	22	10	.350	.625	1	2.380	1.844	1.468
57					16	.625	.937	1	2.380	2.000	1.468
58					20	.875	1.250	2	2.550	2.230	1.468
59		24	24	24	16	.625	.937	1	2.380	2.000	1.593
60			28		12	.500	.750	1	2.550	1.916	1.969

Dash No.	E ±.125	V Thread Class 2A	F Max	G Dia Max	H Max
51	1.916	1.4375-18 UNEF	2.87	1.468	3.26
52	1.016	1.1875-18 UNEF	1.86	-	2.25
53	1.166	1.4375-18 UNEF	2.14	-	2.53
54	1.016	1.1875-18 UNEF	1.86	-	2.25
55	2.016	1.750-18 UNS	3.14	1.969	3.53
56	1.166	1.000-20 UNEF	2.11	-	2.50
57	1.166	1.4375-18 UNEF	2.12	-	2.51
58	2.016	1.750-18 UNS	3.14	1.969	3.53
59	1.166	1.4375-18 UNEF	2.14	-	2.53
60	1.266	1.1875-18 UNEF	2.33	-	2.88

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TABLE I. Dimensions (continued).

Dash No.	Fits Shell of Connector				Clamp Size	A (see note 6) Wire Bundle Accommodation Range		Style	B ±.062	C Ref	D Dia Max
	MIL-C-81703 Ser 3	MIL-C-26482 Ser 2	MIL-C-5015 Crimp	MIL-C-83723 Ser III		Min	Max				
61			28		20	.875	1.250	1	2.550	2.230	1.969
62			32		24	1.000	1.375	1	2.800	2.024	2.219
63			36		20	.875	1.250	1	2.770	2.230	2.469
64					28	1.250	1.625	1	2.770	2.550	2.469
65			40		20	.875	1.250	1	2.770	2.230	2.719
66					28	1.250	1.625	1	2.770	2.550	2.719
67			44		20	.875	1.250	1	2.770	2.230	2.969
68					28	1.250	1.625	1	2.770	2.550	2.969
69			48		20	.875	1.250	1	2.770	2.230	3.219
70					28	1.250	1.625	1	2.770	2.550	3.219

Dash No.	E ±.125	V Thread Class 2A	F Max	G Dia Max	H Max
61	1.266	1.750-18 UNS	2.39	-	2.94
62	1.616	2.000-18 UNS	2.86	-	3.41
63	1.816	1.750-18 UNS	3.13	-	3.68
64	1.816	2.250-16 UN	3.27	-	3.82
65	2.566	1.750-18 UNS	3.88	-	4.43
66	2.566	2.250-16 UN	4.02	-	4.57
67	2.566	1.750-18 UNS	3.88	-	4.43
68	2.566	2.250-16 UN	4.02	-	4.57
69	2.566	1.750-18 UNS	3.88	-	4.43
70	2.566	2.250-16 UN	4.02	-	4.57

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TABLE I. Dimensions (continued).

Dash No.	Fits Shell of Connector				Clamp Size	A (see note 6) Wire Bundle Accommodation Range		Style	B ±.062	C Ref	D Dia Max
	MIL-C-81703 Ser 3	MIL-C-26482 Ser 2	MIL-C-5015 Crimp	MIL-C-83723 Ser III		Min	Max				
71	7	12	12,12S	12	10	.350	.625	2	2.100	1.844	.858
72	27	18	18	18	16	.625	.937	2	2.380	2.000	1.218
73		24	24	24	20	.875	1.250	2	2.550	2.230	1.593
74	12	14	14,14S	14	4	.125	.312	1	2.010	1.544	.984
75	19	16	16,16S	16	4	.125	.312	1	2.100	1.544	1.112

Dash No.	E ±.125	V Thread Class 2A	F Max	G Dia Max	H Max
71	1.686	1.000-20 UNEF	2.42	1.112	2.81
72	1.926	1.4375-18 UNEF	2.88	1.468	3.27
73	2.026	1.750-18 UNS	3.15	1.969	3.54
74	.741	.625-24 UNEF	1.31	-	1.70
75	.926	.625-24 UNEF	1.59	-	1.98

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INCH	mm	INCH	mm	INCH	mm	INCH	mm
0.005	0.13	1.250	31.75	1.860	47.24	2.550	64.77
0.010	0.25	1.266	32.16	1.875	47.63	2.566	65.18
0.020	0.51	1.270	32.26	1.916	48.67	2.600	66.04
0.125	3.18	1.280	32.51	1.918	48.72	2.610	66.29
0.250	6.35	1.310	33.27	1.920	48.77	2.680	68.07
0.312	7.92	1.320	33.53	1.926	48.92	2.719	69.06
0.350	8.89	1.335	33.91	1.940	49.28	2.770	70.36
0.375	9.53	1.345	34.16	1.969	50.01	2.800	71.12
0.437	11.10	1.370	34.80	1.970	50.04	2.810	71.37
0.500	12.70	1.375	34.93	1.980	50.29	2.860	72.64
0.541	13.74	1.380	35.05	2.000	50.80	2.870	72.90
0.575	14.60	1.386	35.20	2.010	51.05	2.880	73.15
0.617	15.67	1.391	35.33	2.016	51.21	2.940	74.67
0.625	15.88	1.4375	36.51	2.024	51.41	2.969	75.41
0.636	16.15	1.468	37.29	2.026	51.46	3.000	76.20
0.641	16.28	1.470	37.34	2.030	51.56	3.060	77.72
0.669	16.99	1.480	37.59	2.050	52.07	3.070	77.98
0.700	17.78	1.544	39.22	2.100	53.34	3.130	79.50
0.734	18.64	1.570	39.87	2.110	53.59	3.140	79.76
0.741	18.82	1.590	40.39	2.120	53.85	3.150	80.01
0.750	19.05	1.593	40.46	2.140	54.36	3.219	81.76
0.779	19.79	1.616	41.05	2.160	54.86	3.260	82.80
0.858	21.79	1.625	41.28	2.195	55.75	3.270	83.06
0.875	22.22	1.653	41.99	2.219	56.36	3.360	85.34
0.904	22.96	1.655	42.04	2.230	56.64	3.390	86.11
0.926	23.52	1.660	42.16	2.250	57.15	3.410	86.61
0.937	23.80	1.670	42.41	2.270	57.66	3.530	89.66
0.984	24.99	1.686	42.82	2.290	58.17	3.540	89.91
1.000	25.40	1.700	43.18	2.360	59.94	3.620	91.94
1.016	25.81	1.745	44.32	2.410	61.21	3.680	93.47
1.029	26.14	1.750	44.45	2.415	61.34	3.800	96.52
1.090	27.69	1.760	44.70	2.420	61.47	3.820	97.02
1.112	28.24	1.766	44.86	2.420	61.46	3.880	98.55
1.144	29.06	1.770	44.96	2.445	62.10	3.940	100.07
1.166	29.62	1.800	45.72	2.469	62.71	4.020	102.11
1.180	29.97	1.816	46.13	2.500	63.50	4.140	105.16
1.184	30.07	1.840	46.74	2.510	63.75	4.430	112.52
1.200	30.48	1.844	46.84	2.530	64.26	4.570	116.07
1.218	30.94						

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REQUIREMENTS:

1. Design and construction:

   a. Dimensions and configuration: See figure 1 and table I.

   b. Interface dimensions shall conform to MIL-C-85049 figure 4.

   c. Figure 1 consists of a coupling nut and follower. The coupling nut shall be captivated to and free to rotate on the follower.

2. Material and finish:

   a. Accessory components shall be as shown in table II.

   b. Grommet and O-Ring - Grommet and o-ring elastomer material shall be in accordance with MIL-C-85049.

   c. Clamp assembly screws - Corrosion resistant steel in accordance with MIL-C-85049, passivated per QQ-P-35.

3. Qualification: See MIL-C-85049, category 1A. Information pertaining to qualification of products may be obtained from the Commanding Officer, Naval Avionics Center, Attention: Code B/444, 6000 East 21st Street, Indianapolis, IN 46219-2189.

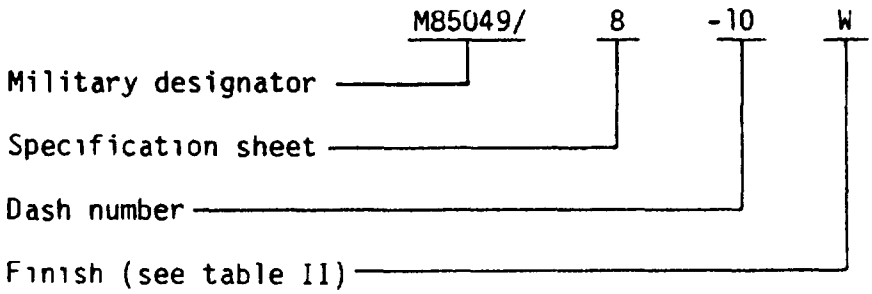
4. Applications:

   a. These accessories are for shield termination and environmental applications. Armor shield termination and environmental applications may be achieved by reversing the ferrules and grommets.

   b. Style 2 applies when the cable diameter exceeds the "E" diameter of the interface dimensions defined in MIL-C-85049 figure 4.

   c. For assembly instructions see figure 2 of MIL-C-85049/10.

5. Part Number: See example below:



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6. Supersession data:

MS3188A * * M85049/8 - * *

Dash Number

Finish

Old	New
A	W
C	W

TABLE II. Material and finish.

Component	Material	Finish
Body	Aluminum alloy in accordance with MIL-C-85049	1/ N, 2/ W
Body	Corrosion resistant steel in accordance with MIL-C-85049	B, S

1/ Not for Navy use. Restricted to Air Force Space applications only.
2/ W finish is not for use in space application.

NOTES:

1. Dimensions are in inches.
2. Metric equivalents shall be based upon 1.00 inch = 25.4 mm and shall be in accordance with ASTM E380, paragraph 4.5, Method A and ASTM E29, rounding-off method.
3. Unless otherwise specified, tolerances shall be: .XX = ±.01; .XXX ±.005; angular; X° = ± 1°; X°XX' = ±0°30'.
4. Dimensions apply after plating.
5. Loosely assembled is defined as the MIL-C-85049/42 clamp tightened to the MIL-C-85049/8 connector accessory body a minimum of two full thread lengths.
6. "Wire bundle accommodation range" is defined as the envelope area for the wire bundle. This dimension is not meant to define the clamp hardware limits.

Revision letters are not used to denote changes due to the extensiveness of the changes.

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